

Harshal Donarkar

AI Engineer — Deep Learning · LLMs/RAG · Computer Vision · Production ML

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AI Engineer specializing in LLMs, RAG, and computer vision, with production deployments and two peer-reviewed publications. Experience shipping multimodal AI pipelines, optimizing inference latency, and building systems end-to-end — from model training to containerized deployment.

EXPERIENCE

AI Engineer Intern

Sambhav AI and Tech Services · Nagpur

- Deployed AI video processing using Whisper, Tesseract OCR & EasyOCR; containerized with Docker & Kubernetes — cut processing time by 50%.
- Integrated multimodal AI models with an EdTech platform for auto-generated topic summaries and enhanced learning.
- Optimized AI search via offline processing + cloud-based uploads — increased query speed by 75%.
- Revamped the deployment pipeline with automated testing & monitoring — boosted model accuracy by 10%.

Feb 2025 – Aug 2025

PROJECTS

Vault — Local-First Encrypted Personal Data Platform

- Built a fully offline, zero-network personal data platform: Rust core (Tauri 2) + React/TypeScript UI communicating only through a locked-down typed IPC boundary.
- Designed the security model end-to-end — SQLCipher AES-256 full-database encryption, Argon2id key derivation, automatic key wiping, 10-minute auto-lock, and leak-free error handling.
- Built resilient parsers for real-world bank statement formats (SBI CSV, Jupiter XLSX) with UPI merchant-name extraction and a retroactive, reversible rule-based categorization engine.

2026

Healthcare Recommendation System [\[GitHub\]](#)

- Built a multimodal clinical decision-support system combining BERT-based symptom diagnosis, chest X-ray screening, and OCR-based lab report analysis (Flask + PyTorch), with natural-language explanations via 5 integrated LLM providers.
- Fine-tuned DistilBERT/PubMedBERT on the DDXPlus benchmark (1M+ simulated patient cases, 49 pathologies) using KL-divergence soft-label training, cutting probability miscalibration 15x and improving NDCG@3 from 0.748 to 0.921 versus a TF-IDF baseline; covered by 67 automated tests.

2025

ConstructionMartt CRM [\[Live\]](#)

- Built and deployed a full-stack CRM (FastAPI, MongoDB, React Native/Expo PWA) for a construction materials trading business — live and used daily by the team.
- Implemented role-based access (Admin/Sales/Accounts/Operations), real-time dashboard analytics, PDF/Excel document generation, and a 12-stage construction-site tracking pipeline for demand forecasting.

Dec 2025 – Present

DevBridge — AI Pre-Dev Automation [\[GitHub\]](#)

- Built a multimodal pipeline using GPT-4 and Claude in parallel to auto-generate project documentation with D3.js dependency visualizations, cutting documentation time by 65%.

Apr 2025

MCP Server for Node.js Automation [\[GitHub\]](#)

- Built a multi-tool MCP server (Python/FastMCP) exposing 10+ tools for Node.js project automation — scaffolding, TypeScript migration, Docker setup, and live npm registry integration — built 5 months after the MCP spec's release.

Apr 2025

Multimodal AI Video Summarizer

- Built a timestamp-indexed search system across educational videos using Whisper transcription and Tesseract/EasyOCR slide extraction; containerized with Docker & Kubernetes; cut search time by 30% across 20 multilingual videos.

Mar 2025

Skin Cancer Prediction Model — Published, IJCNIS [\[GitHub\]](#)

- Trained a CNN on a combined HAM10000 + ISIC 2019 dataset (35K+ dermoscopic images) classifying 7 lesion types, achieving 80%+ TPR; addressed class imbalance via oversampling (to ~52K) and augmentation; deployed as a Flask/Streamlit app for real-time screening.

Aug 2024

Breast Cancer Prediction System

- Built a classifier on 10,000+ patient records achieving 97% accuracy; reduced data skewness by 18% through advanced normalization techniques.

Feb 2024

Optimized Job Search Engine — Published, IRJET

- Built a Selenium/BeautifulSoup scraper with layered filters (location, skills, experience, salary); improved pipeline data efficiency by 15%.

Dec 2023

PUBLICATIONS

"Skin Cancer Prediction Model using Deep Learning" — IJCNIS, Vol 16, Issue 5, 2024 (ISSN 2073-607X)
"Optimising Job Search using Web Scraping" — IRJET, Vol 11, Issue 4, April 2024 (Impact Factor 8.226)

SKILLS

ML / Deep Learning	TensorFlow, PyTorch, Keras, Scikit-Learn, CNNs, Transformers
LLMs & AI Eng.	Claude (primary — architecture, code review, debugging), GPT-5.5, Gemini 3.1, Llama 4, Mistral, Groq, LangChain, RAG, Vector DBs (Pinecone, ChromaDB), Prompt Engineering, Agents, Function Calling
CV / NLP	OpenCV, Whisper, OCR (Tesseract, EasyOCR)
Backend / Full-stack	Python, Node.js, Flask, FastAPI, REST, TypeScript, Next.js, Rust (Tauri)
MLOps / Infra	Docker, Kubernetes, CI/CD, HuggingFace, Cloudflare Workers
Databases	MongoDB, MySQL, PostgreSQL, Pinecone, ChromaDB, SQLCipher

EDUCATION

B.Tech, Computer Science & Engineering — Rashtrasant Tukdoji Maharaj Nagpur University

2021 – 2025 · CGPA 7.74

ACHIEVEMENTS & LEADERSHIP

2nd Runner-Up — 'Hackengers' Hackathon, Livewire Teaching Institute, Chennai (2024)
Technical Head, Coding Club, KDKCE — mentored 30+ students in deep learning, Python & ML workflows (2023–24)